STATE OF NEW MEXICO
OFFICE OF THE STATE ENGINEER
Hydrology Bureau

MISCELLANEOUS MEMORANDUM

DATE: May 24, 2024

TO: Ramona Martinez, District VI Manager, Water Rights Division
Lorraine Garcia, Upper Rio Grande Basin Manager, Water Rights Division
Cindy Stokes, Interstate Stream Commission

THROUGH: Katie Zemlick, Chief, Hydrology Bureau, Santa Fe, NM KZ

FROM: Brad Wolaver, Senior Hydrologist, Hydrology Bureau, Santa Fe, NM BDW
Laura Petronis, Water Resources Manager 1, Hydrology Bureau, Santa Fe, NM LPP

SUBJECT: Calendar Year 2023 hydrologic impacts on the Rio Grande, Rio Pojoaque-Nambe, Rio Tesuque, and La Cienega springs due to Buckman well field pumping, RG-20516 et al.

KEYWORD: Miscellaneous Memorandum, District VI, Santa Fe County, OSE Rio Grande Basin, City of Santa Fe

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SUMMARY AND CONCLUSION

This memorandum describes the estimated effects on the Rio Grande, Rio Pojoaque-Nambe, and Rio Tesuque due to calendar year (CY) 1972 through 2023 from Buckman well field pumping and the estimated effects on La Cienega springs due to CY1988 through CY2023 Buckman well field pumping.

CY1988 to CY2023 diversions from the Buckman well field reported by the City of Santa Fe are presented in Table 1. Monthly CY2023 Buckman diversions reported by the City of Santa Fe are presented in Table 2. CY2023 pumping totaled 866.48 acre-feet (ac-ft)[1], which represents the eighth lowest annual pumping from the Buckman well field since 1988 (the earliest year considered in Table 1). Newer wells 10 through 13 supplied 382.27 ac-ft (44.1 percent) of the 2023 diversion from the Buckman well field. The three wells (Buckman 1, 7, and 8) in the model cell closest to the Rio Grande (row, column: 13, 11) supplied a total of 349.50 ac-ft (40.3 percent) of the 2023 Buckman well field diversion.

[1] CY2023 Buckman well field pumping reported by the City of Santa Fe was 866.46 ac-ft; the 0.02 ac-ft difference is thought to be due to rounding.

SIMULATED STREAM DEPLETIONS ON THE RIO GRANDE AND SANTA FE-AREA TRIBUTARIES

Streamflow depletion on the Rio Grande, Rio Pojoaque-Nambe, and Rio Tesuque due to Buckman well field pumping from CY1972 to CY2023 was estimated by following the general procedure described in Compton (1988). It specifies that residual hydrologic effects on the Rio Grande and its tributaries resulting from CY1972 through CY1987 Buckman well field pumping be calculated using the NMOSE analytical model, and that post-1987 pumping effects on nearby streams be calculated using the McAda and Wasiolek (1988) ground-water flow model.

Residual hydrologic effects on the Rio Pojoaque-Nambe and Rio Tesuque due to CY1972-CY1987 Buckman well field pumping were calculated for CY1988 through CY2010 using the NMOSE analytical model and described in Spinks (1988). Core (2003) projected CY1972-CY1987 Buckman well field pumping effects on the Rio Tesuque and Rio Pojoaque-Nambe after CY2010. Residual hydrologic effects on the Rio Grande due to CY1972-CY1987 Buckman well field pumping were calculated using the NMOSE analytical model and described in Fleming (1989).

Prior to 2004, the McAda-Wasiolek (1988) model was used to calculate impacts for 1988 through 2003 (based on pumping from CY1988 through CY2003). For the 2004 calculation (based on pumping from CY1988 through CY2004), and for subsequent annual calculations, the revised superposition version of the McAda-Wasiolek (1988) model (Core, 1996, Barroll and Keyes, 2005) was used and was run with MODFLOW96. Use of the superposition model has consolidated the number of separate model runs needed, and additionally has allowed for estimation of effects at La Cienega springs (see below). The total diversion for each of the Buckman wells was divided equally between model layers 1 and 2 and monthly pumping was simulated in the model, which is consistent with how pumping has been simulated in previous years. For this analysis, CY2023 monthly pumping from the Buckman wells was added to the well file (which contained the previously entered CY1988 though CY2022 pumping), the model was run, and the results were processed.

Table 3 summarizes the estimated streamflow depletion on the Rio Pojoaque-Nambe and Rio Tesuque from 1988 to 2025 due to Buckman well field pumping from CY1972 to CY2023. CY2023 stream depletion was calculated to be 60.323 ac-ft on the Rio Pojoaque-Nambe system, and 33.491 ac-ft on the Rio Tesuque. Starting in CY2023, maximum calendar year depletions based on Buckman well field pumping from CY1972 to CY2023 were estimated to be 60.683 ac-ft (estimated to occur in CY2024) and 33.572 ac-ft (estimated to have occurred in CY2024), in the Rio Pojoaque-Nambe system and the Rio Tesuque, respectively.

Table 4 summarizes the monthly estimated streamflow depletion on the Rio Grande above and below Otowi gage for CY2023. For CY2023, the total estimated streamflow depletion on the Rio Grande for the reach above Otowi gage is 102.65 ac-ft and for the reach below Otowi gage is 738.23 ac-ft.

SIMULATED FLOW REDUCTION FROM LA CIENEGA SPRINGS

The effect of Buckman well field pumping on La Cienega spring flow was calculated using the revised superposition version of the McAda-Wasiolek (1988) model (Core, 1996, Barroll and Keyes, 2005). The total diversion for each of the Buckman wells was divided equally between model layers 1 and 2 and monthly pumping was simulated in the model.

Table 5 summarizes the estimated depletion to La Cienega springs from 2004 to 2025 due to Buckman well field pumping from CY1988 to CY2023. CY1988 to CY2023 Buckman well field pumping is estimated to have caused a spring flow reduction at La Cienega of 3.54 ac-ft during CY2023. Based on pumping during this period, the hydrologic effect on the La Cienega Springs would reach 16.51 ac-ft in CY2152. It is apparent that a significant lag occurs between timing of actual Buckman pumping and the incursion of simulated maximum effect on flow at the La Cienega springs.

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TABLES

Table 1. Reported Buckman Well Field pumping in acre-feet

Year	Well B1	Well B2	Well B3/3A	Well B4	Well B5	Well B6	Well B7	Well B8	Well B9	Well B10	Well B11	Well B12	Well B13	ANNUAL TOTAL
1988	547.85	742.16	82.49	465.34	52.44	581.03								2471.31
1989	563.10	1271.65	260.73	1049.03	146.30	1259.71								4550.52
1990	333.83	840.18	263.18	1041.18	142.50	1202.73								3823.60
1991	200.79	544.79	177.28	625.51	109.60	801.59	301.83	377.90						3139.31
1992	194.44	729.85	112.18	497.37	48.19	1215.33	993.01	961.78						4752.13
1993	692.78	735.49	137.31	647.47	140.61	1222.32	1070.03	962.28						5608.29
1994	707.94	743.24	125.12	667.62	242.34	1051.71	638.06	777.44						4953.47
1995	945.40	824.76	351.55	765.18	384.14	1009.37	861.37	778.63						5920.39
1996	935.43	707.33	616.53	631.95	437.94	972.46	592.58	761.45						5655.67
1997	924.65	815.53	467.03	453.35	139.11	1030.86	497.11	347.11						4674.75
1998	660.42	878.84	457.50	399.05	110.57	1042.80	742.71	923.63						5215.52
1999	624.75	841.62	415.54	353.42	0.00	993.70	1159.01	890.50						5278.54
2000	626.28	720.21	519.30	266.50	0.00	914.97	1169.10	864.24						5080.60
2001	716.03	489.80	436.06	354.01	0.00	1037.41	1032.51	677.52						4743.34
2002	730.45	583.98	416.69	554.62	265.43	1272.77	1145.18	868.53						5837.66
2003	508.17	540.25	337.84	386.17	234.42	1187.87	987.07	780.44	292.08	0.03	251.23	259.45	60.07	5825.09
2004	26.51	6.17	73.44	355.22	3.76	958.14	277.25	652.62	5.67	453.64	710.82	948.33	1265.27	5736.85
2005	4.99	6.39	105.14	510.34	1.04	367.40	1223.04	797.05	1.73	22.76	367.91	216.31	160.25	3784.34
2006	20.00	150.23	247.19	612.03	7.66	1076.65	237.57	814.32	54.37	656.51	824.75	88.36	407.43	5197.06
2007	398.50	8.63	183.29	565.27	0.30	408.79	703.51	704.39	25.12	222.53	136.92	31.49	404.83	3793.58
2008	75.49	0.00	115.27	782.12	60.02	37.57	851.48	866.44	0.00	262.74	556.46	42.16	84.28	3734.04
2009	29.56	0.00	143.06	704.14	157.32	951.76	530.90	367.97	0.00	233.39	330.98	345.38	7.89	3802.35
2010	81.60	0.00	437.16	474.20	128.67	1023.46	249.29	7.70	0.00	96.65	328.05	181.80	44.33	3052.91
2011	77.93	0.00	101.38	219.65	0.62	432.63	0.00	244.35	0.00	194.11	288.53	165.30	0.01	1724.51

Table 1. Reported Buckman Well Field pumping in acre-feet (cont.)

Year	Well B1	Well B2	Well B3/3A	Well B4	Well B5	Well B6	Well B7	Well B8	Well B9	Well B10	Well B11	Well B12	Well B13	ANNUAL TOTAL
2012	213.82	0.00	95.35	149.52	48.83	316.83	0.00	75.33	0.00	20.09	0.00	130.64	0.00	1050.42
2013	631.14	4.60	218.81	403.80	0.00	514.24	31.48	255.14	4.85	184.98	288.29	159.11	194.03	2890.47
2014	54.03	0.00	81.58	33.97	29.46	134.59	53.67	20.63	70.81	0.00	29.67	40.50	25.89	574.80
2015	163.55	0.00	56.61	11.98	14.31	163.26	32.89	49.61	7.74	0.01	25.21	29.26	73.34	627.77
2016	231.96	0.00	37.16	85.96	0.01	220.13	89.25	150.83	19.23	28.69	4.41	29.59	27.60	924.82
2017	335.63	0.00	0.01	29.75	0.00	224.68	0.20	67.52	0.04	0.05	5.28	0.00	10.93	674.10
2018	361.44	0.01	16.94	44.58	0.00	160.71	0.10	55.57	14.39	33.94	40.59	14.41	88.48	831.17
2019	95.14	0.15	21.36	16.83	0.74	90.59	1.45	64.00	1.03	13.20	0.03	0.00	9.57	314.09
2020	276.18	0.00	6.64	25.38	0.02	111.77	0.09	127.51	1.82	48.51	15.68	3.20	15.80	632.60
2021	241.72	0.00	3.85	35.38	0.00	74.29	12.25	98.06	25.85	32.08	0.00	10.97	124.48	658.93
2022	458.14	0.01	0.40	38.18	0.00	7.92	0.31	196.50	11.81	45.17	89.00	0.18	127.85	975.47
2023	224.06	0.00	50.15	47.36	0.00	31.36	0.37	125.08	5.83	4.65	0.17	0.14	377.31	866.48

Average annual pumpage, 1988 (or start-up) through 2023	386.49	338.50	199.20	397.32	80.73	669.54	469.23	476.12	25.83	121.61	204.47	128.41	167.13	3316.03
Most-recent 3-year average annual pumpage	307.97	0.00	18.13	40.31	0.00	37.85	4.31	139.88	14.50	27.30	29.72	3.76	209.88	833.63

Table 2. CY2023 monthly pumping from the Buckman well field (in acre-feet)

WELL NUMBER	JAN 2023	FEB 2023	MAR 2023	APR 2023	MAY 2023	JUN 2023	JUL 2023	AUG 2023	SEP 2023	OCT 2023	NOV 2023	DEC 2023	TOTAL 2023
Buckman #1	0.15	40.76	0.46	45.28	63.38	6.96	13.63	16.80	35.46	0.37	0.68	0.14	224.06
Buckman #2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Buckman #3A	0.00	0.00	0.06	0.05	30.53	10.45	9.07	0.00	0.00	0.00	0.00	0.00	50.15
Buckman #4	0.08	0.00	0.09	0.09	36.66	10.44	0.00	0.00	0.00	0.00	0.00	0.00	47.36
Buckman #5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Buckman #6	0.12	0.00	0.29	5.87	5.91	8.92	7.86	0.14	1.86	0.17	0.11	0.10	31.36
Buckman #7	0.00	0.00	0.00	0.10	0.00	0.01	0.02	0.08	0.08	0.00	0.07	0.00	0.37
Buckman #8	0.10	26.08	0.01	3.95	0.42	0.37	14.24	23.05	20.23	0.09	0.48	36.06	125.08
Buckman #9	0.07	0.00	0.14	0.09	2.29	2.82	0.06	0.09	0.14	0.00	0.06	0.06	5.83
Buckman #10	0.35	0.25	0.08	0.09	2.95	0.23	0.00	0.27	0.12	0.09	0.14	0.09	4.65
Buckman #11	0.00	0.00	0.02	0.00	0.00	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.17
Buckman #12	0.00	0.00	0.00	0.00	0.00	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.14
Buckman #13	0.17	0.00	0.27	0.10	61.06	140.19	140.85	34.67	0.00	0.00	0.00	0.00	377.31
SUM	1.03	67.10	1.42	55.62	203.18	180.68	185.74	75.10	57.90	0.72	1.54	36.44	866.48

Table 3. Estimated impacts from reported Buckman Well Field pumping through Calendar Year 2023

Year	Residual impact (acre-feet) of 1972 - 1987 Buckman pumping (a) (Analytical Model)	Impact (acre-feet) of 1988 - 2023 reported Buckman pumping (b) [no pumping after 2023] 3D Digital Model (1988-2003) (c) Revised Superposition Model (2004-current) (d)	Total impact (acre-feet)
1988	**40.432**	1.060	41.492
1989	**39.244**	3.287	42.531
1990	**37.971**	5.480	43.451
1991	**36.557**	7.319	43.876
1992	**34.928**	9.960	44.888
1993	**33.112**	13.118	46.230
1994	**31.185**	15.942	47.127
1995	**29.226**	18.910	48.136
1996	**27.296**	21.873	49.169
1997	**25.439**	24.144	49.583
1998	**23.678**	26.656	50.334
1999	**22.028**	29.379	51.407
2000	**20.491**	32.254	52.745
2001	**19.068**	34.714	53.782
2002	**17.753**	37.610	55.363
2003	**16.543**	40.578	57.121
2004	**15.429**	41.763	57.192
2005	**14.404**	42.778** (42.781)	57.182** (57.185)
2006	**13.462**	44.297** (44.306)	57.759** (57.768)
2007	**12.595**	46.327	58.922
2008	**11.797**	48.426	60.223
2009	**11.061**	50.005	61.066
2010	**10.383**	51.723	62.106
2011	6.151	53.077	59.228
2012	4.693	53.850	58.543
2013	3.234	54.410	57.644
2014	1.775	55.726	57.501
2015	0.316	56.351	56.667
2016	(e)	56.944	56.944
2017		57.520	57.520
2018		58.133	58.133
2019		58.683	58.683
2020		59.154	59.154
2021		59.395	59.395
2022		59.844	59.844
2023		60.323	60.323
2024		60.683	60.683
2025		60.319	60.319

Year	Residual impact (acre-feet) of 1972 - 1987 Buckman pumping (a) (Analytical Model)	Impact (acre-feet) of 1988 - 2023 reported Buckman pumping (b) [no pumping after 2023] 3D Digital Model (1988-2003) (c) Revised Superposition Model (2004-current) (d)	Total impact (acre-feet)
1988	**21.015**	0.145	21.160
1989	**22.333**	0.601	22.934
1990	**23.391**	1.166	24.557
1991	**24.227**	1.629	25.856
1992	**24.868**	2.163	27.031
1993	**25.327**	2.867	28.194
1994	**25.615**	3.497	29.112
1995	**25.747**	4.279	30.026
1996	**25.737**	4.929	30.666
1997	**25.608**	5.517	31.125
1998	**25.378**	6.154	31.532
1999	**25.067**	6.820	31.887
2000	**24.691**	7.412	32.103
2001	**24.265**	7.956	32.221
2002	**23.8**	8.659	32.459
2003	**23.308**	9.506	32.814
2004	**22.797**	10.222	33.019
2005	**22.273**	11.294** (11.295)	33.567** (33.568)
2006	**21.743**	11.967** (11.968)	33.680* (33.711)
2007	**21.212**	12.709	33.921
2008	**20.683**	13.363	34.046
2009	**20.157**	13.875	34.032
2010	**19.639**	14.526	34.165
2011	19.258	15.163	34.421
2012	18.767	15.633	34.400
2013	18.276	15.946	34.222
2014	17.785	16.474	34.259
2015	17.295	16.890	34.185
2016	16.804	17.256	34.060
2017	16.313	17.588	33.901
2018	15.822	17.974	33.796
2019	15.331	18.385	33.716
2020	14.841	18.831	33.672
2021	14.35	19.180	33.530
2022	13.859	19.631	33.490
2023	13.368	20.123	33.491
2024	12.877	20.695	33.572
2025	12.387	21.018	33.405
	(f)		

(a) Data from: Compton (1989) and Core (2003).
(b) Model-predicted streamflow depletion due to Buckman wellfield pumping CY1988–2023 was calculated using model cell placements for Buckman wells 5 and 8 based on 4/28/2004 GPS measurements.
(c) The McAda and Wasiolek (1988) groundwater flow model was used to calculate impacts for CY1988–2003 due to simulated Buckman wellfield pumping from CY1988–2003.
(d) The Revised Superposition Model (Core, Andrew B., 1996; Barroll, Peggy and Keyes, Eric, 2005) was used to calculate impacts starting in CY2004 due to simulated Buckman wellfield pumping from CY1988–2023.
(e) Residual impact of CY1972–87 Buckman pumping on the Rio Pojoaque-Nambe estimated at near nil by CY2017 (Core, Andrew B., 2003).
(f) Residual impact of CY1972–87 Buckman pumping on the Rio Tesuque estimated at near nil by CY2050 (Core, Andrew B., 2003).
** Value prior to correction of simulated pumping rates in February 2005 and February 2006 (corrected value shown in parenthesis for 2005 and 2006).

Table 4. Estimated CY2023 monthly effects on the Rio Grande due to CY1972–2023 (in acre-feet) Buckman well field pumping

	Jan-23	Feb-23	Mar-23	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23	Total 2023
ABOVE OTOWI:													
Revised Superposition Model (pumping 1988-2021) [a]	8.79	7.93	8.76	8.46	8.73	8.44	8.71	8.70	8.41	8.68	8.39	8.65	102.65
BELOW OTOWI													
Revised Superposition Model (1988-2021) [a]	63.75	61.74	62.02	62.03	67.92	61.12	62.12	62.89	63.66	60.01	53.91	57.12	738.28

(a) Depletions due to CY1972–1987 Buckman well field pumping calculated using the Analytical Model are 0 ac-ft in CY2023 based on data in Fleming (1989).

Table 5. Estimated impacts on La Cienega springs due to Buckman Well Field pumping through Calendar Year 2023

Year	Impact (acre-feet) of 1988 - 2023 reported Buckman pumping (a) [no pumping after 2023] using the Revised Superposition Model (b)
2004	0.45
2005	0.66
2006	0.83
2007	0.99
2008	1.16
2009	1.32
2010	1.49
2011	1.65
2012	1.82
2013	1.97
2014	2.13
2015	2.29
2016	2.45
2017	2.60
2018	2.75
2019	2.90
2020	3.06
2021	3.21
2022	3.37
2023	3.54
2024	3.74
2025	3.90

(a) Model-predicted streamflow depletion due to CY1988–2023 Buckman Well Field pumping was calculated using model cell placements for Buckman wells 5 and 8 based on 4/28/2004 GPS measurements.

(b) The Revised Superposition Model (Core, 1996, Barroll and Keyes, 2005) was used to calculate impacts due to simulated CY1988–2023 Buckman wellfield pumping.